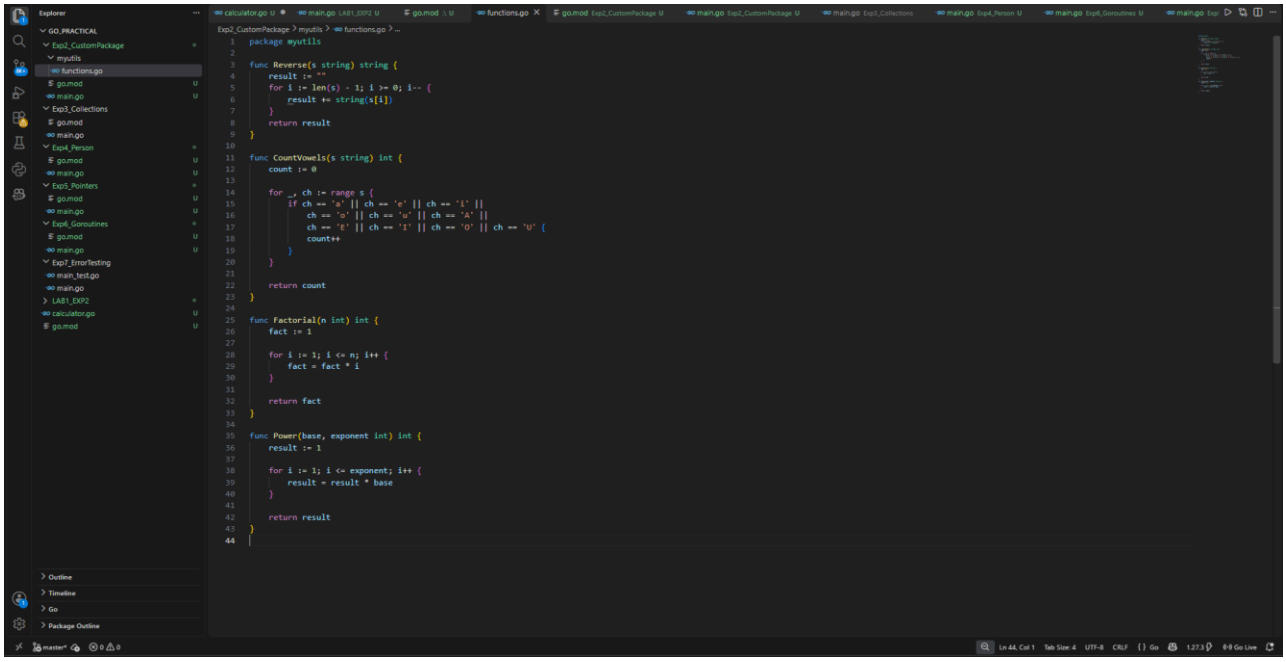
EXP 2

1. Custom Reusable Package with String and Math Functions

    1. Create a custom Go package containing at least two string-manipulation functions (e.g., Reverse, CountVowels) and two mathematical utility functions (e.g., Factorial, Power).
    2. Exported names must begin with an uppercase letter.
    3. Call all four functions from a separate main. Go program to demonstrate package import and reusability.
    4. Push the complete package directory and main program to your GitHub repository.
    5. Upload a PDF/image file of the output on the portal.



```go
package myutils

func Reverse(s string) string {
    result := ""
    for i := len(s) - 1; i >= 0; i-- {
        result += string(s[i])
    }
    return result
}

func CountVowels(s string) int {
    count := 0

    for _, ch := range s {
        if ch == 'a' || ch == 'e' || ch == 'i' ||
            ch == 'o' || ch == 'u' || ch == 'A' ||
            ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U' {
            count++
        }
    }

    return count
}

func Factorial(n int) int {
    fact := 1

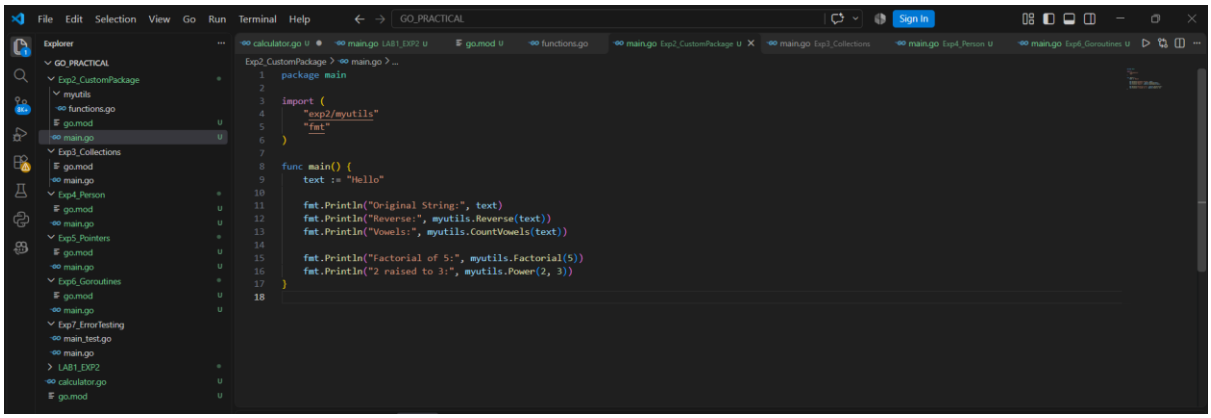
    for i := 1; i <= n; i++ {
        fact = fact * i
    }

    return fact
}

func Power(base, exponent int) int {
    result := 1

    for i := 1; i <= exponent; i++ {
        result = result * base
    }

    return result
}
```



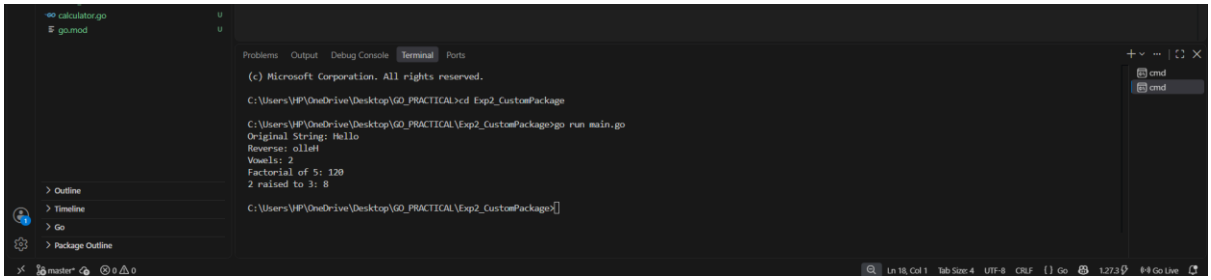
```go
package main

import (
    "exp2/myutils"
    "fmt"
)

func main() {
    text := "Hello"

    fmt.Println("Original String:", text)
    fmt.Println("Reverse:", myutils.Reverse(text))
    fmt.Println("Vowels:", myutils.CountVowels(text))

    fmt.Println("Factorial of 5:", myutils.Factorial(5))
    fmt.Println("2 raised to 3:", myutils.Power(2, 3))
}
```


calculator.go
go.mod
Problems
Output
Debug Console
Terminal
Ports
(c) Microsoft Corporation. All rights reserved.
C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL>cd Exp2_CustomPackage
C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL\Exp2_CustomPackage>go run main.go
Original String: Hello
Reverse: olleH
Vowels: 2
Factorial of 5: 120
2 raised to 3: 8
C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL\Exp2_CustomPackage>
cmd
cmd
Outline
Timeline
Go
Package Outline
master*
Ln 18, Col 1
Tab Size: 4
UTF-8
CRLF
Go
1.27.3
Go Live